Subhransu S. Bhattacharjee

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Professional Summary

Target roles: Data Scientist; Applied Scientist; Research Scientist; Research Engineer; Quantitative Researcher; Quantitative Developer.
Bio: AI PhD candidate at the Australian National University researching computer vision, generative world models, multimodal representation learning, and token-efficient LLM/VLM architectures for spatial reasoning and efficient inference. First author of papers at CVPR 2025 and ECCV 2026, with applied AI, model evaluation, information retrieval, and quantitative machine learning experience at Microsoft, Optiver, and JPMorgan.

Education

Doctor of Philosophy — **Artificial Intelligence**
School of Computing, Australian National University, Australia — *Apr 2024 – Mar 2027 (expected)*
- **Thesis topic:** *Into the Unknown: Tackling Spatio-semantic Uncertainty using Generative Posteriors*
- **Supervisors:** Dr. Rahul Shome, Dr. Dylan Campbell, and Prof. Stephen Gould

Master of Philosophy in Computer Science *(transferred to PhD)*, *School of Computing, ANU* — *Apr 2023 – Apr 2024*

Bachelor of Engineering (Honours) — **Mechatronics**
School of Engineering, Australian National University, Australia — *Jul 2018 – Dec 2022*
- **First Class Honours**; GPA 6.48/7.00; ranked 3rd in cohort. **Minors:** Mathematics; Electronic Communication Systems.
- **Machine Learning**, London School of Economics Summer School — **Grade A**.
- **Thesis project:** Whiplash Gradient Descent Dynamics.

Professional Experience

Microsoft Corporation — **May 2026 – Jul 2026**
Applied Scientist PhD Research Intern, LLM-as-Judge, RAG & Agent Evaluation
- Built an end-to-end LLM-as-judge framework for RAG and agent outputs in collaboration with Microsoft India Development Center, Microsoft Research India, and Microsoft AI (MAI), combining groundedness, relevance-purity, response-discipline, and dissatisfaction-aware evaluation.
- Built an agentic ingestion harness for unstructured Likert annotations, reducing annotator cognitive load by ∼35% and doubling usable throughput for golden-set construction, evaluator calibration, and failure analysis.

Optiver APAC — **Nov 2024 – Feb 2025**
Quantitative Research Intern, Neural–Fuzzy Modelling, xLSTM, XGBoost, SHAP
- Developed a neural–fuzzy joint classifier for Korean retail-flow prediction, achieving 94% classification accuracy while integrating xLSTM and XGBoost within a streaming market-feature pipeline with online SHAP attribution.
- Built leakage-aware walk-forward evaluation and model-monitoring infrastructure for Korean market microstructure, and separately developed and backtested a statistical-arbitrage model for the Hong Kong market.

JPMorgan Chase & Co. — **Dec 2021 – Feb 2022**
Quantitative Research Intern, NLP, Information Retrieval, Risk
- Developed a metadata-indexing and retrieval pipeline over 10,000+ financial documents supporting transformer-based question answering and structured answer extraction for risk-research workflows, improving top-k retrieval recall by ∼8 percentage points.

Selected Publications

Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome: *Believing is Seeing: Unobserved Object Detection using Generative Models. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, Nashville, Tennessee, USA. Project Page.
Defines 2D, 2.5D, and 3D unobserved object detection from a single RGB view with diffusion/VLM pipelines and tailored metrics; the 3D-diffusion method outperforms all applicable methods, including an in-frustum oracle, on the 3D task.

Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome: *FlatLands: Generative Floormap Completion From a Single Egocentric View. European Conference on Computer Vision (ECCV)*, 2026, Malmö, Sweden. Springer. Project Page.
Releases FlatLands: 270,575 observations from 17,656 real metric indoor scenes across six datasets.
Compares 11 methods, showing stochastic generators improve completion fidelity and capture layout uncertainty.

Subhransu S. Bhattacharjee, Hao Lu, Dylan Campbell & Rahul Shome: *MatterDoor: Sampling Zero-shot Spatio-semantic Priors using Generative Models. arXiv:2510.11014*, 2025. Under review; RSS 2026 FM4RoboPlan workshop presentation, Sydney. Preprint | RSS Workshop Paper.
Samples environment priors from partial observations using pretrained generative models.
Produces RGB-D point-cloud hypotheses for occupancy, target semantics, and configuration-space planning.

Subhransu S. Bhattacharjee & Ian Petersen: *Analysis of the Whiplash Gradient Descent Dynamics. Asian Journal of Control* (Special Issue), Wiley, 2023. DOI: 10.1002/asjc.3153.
Introduces a closed-loop inertial optimiser with adaptive damping, deterministic saddle-point escape, and symplectic Lyapunov convergence analysis. Establishes polynomial and exponential convergence rates for adaptive inertial gradient dynamics. Code.
Conference papers: ASCC 2022, Jeju Island (DOI); ANZCC 2021, Gold Coast (DOI).

Open-Source Research & Software

TorchKAN — Independent PyTorch implementation of Kolmogorov–Arnold Networks with spline, Legendre (KAL-Net), Chebyshev (KAC-Net), and convolutional (KANvolver) variants; classification and regression experiments, CUDA execution, post-training quantisation, and Integrated-Gradients interpretability. Sole maintainer. Project Site | Code

Mangroves — Python package for depth-based hierarchical data management: type-aware ingestion, dynamic attribute access, result caching, and CPU–GPU tensor movement for episodic training pipelines. Sole author; maintenance-only. Project Site | Code

TuringTree — Vectorless, reasoning-based RAG running fully on-device (Ollama + Qwen) with 0–100 confidence scoring and abstention; tests, CI, and a tagged release. Co-developed by a four-person team during the Microsoft Global Intern Hackathon. Project Site | Code

Research Experience

Research School of Management, Australian National University **Sep 2023 – Sep 2024**
Graduate Research Assistant — FinTech & AI, PI: Dr Priya Muthukannan
- Developed an AI-adoption assessment framework for an industry-linked Commonwealth Bank of Australia project.

CIICADA Lab, Australian National University **Dec 2022 – Feb 2023**
Summer Research Scholar — Control & Optimisation, Supervisor: Prof Ian Petersen
- Advanced feedback-based optimisation for systems without closed-form solutions, contributing to ASCC and ANZCC papers.

School of Computing, Australian National University **Mar 2022 – Jun 2022**
Undergraduate Research Scholar — Geometry of Data, Supervisor: Prof Richard Hartley
- Tested invertibility of differentiable neural mappings with dense positional encoding, achieving a 72% hit rate under an RMSE criterion.

Decimal Point Analytics **Dec 2020 – Jan 2021**
NLP Researcher — Financial NLP, India
- Built a RoBERTa-ready financial metadata database and supported client-facing product quality assurance and reassessment.

Armament Research & Development Establishment (ARDE) **May 2020 – Jul 2020**
Summer Research Trainee — Passive Radar Signal Processing, Pune, India
- Developed Kalman-filter-based matched-filter selection for noisy passive-radar signals and an FPGA multiprocessor interface for real-time long-range radar analysis at India's DRDO.

Awards & Recognition

2026 DARPA Travel Grant (AUD 6,000) for the European Conference on Computer Vision (ECCV 2026).
2026 Research contributor on a Google Research grant supporting ANU Robotics work on world models for task-and-motion planning.
2025 Vice-Chancellor's Travel Grant, Australian National University, for CVPR 2025 travel.
2024 Optiver PhD Quant Lab Program (selected participant).
2023 ANU International University Research Scholarship (4 years) and Higher Degree by Research Merit Stipend.
2022 Highly Recommended Paper at the *Asian Control Conference*; Best Student Reviewer Award.
2021 High Commendation Award for undergraduate student paper at the *Australia & New Zealand Control Conference*.
2020 ANU Chancellor's International Scholarship — undergraduate tuition-fee award for academic merit.

Selected Invitations & Talks

2026 Invited attendee at the Gemma release hosted by Google DeepMind, Hyderabad.
2026 Selected invitee to the Hugging Face, PyTorch, and Red Hat Meetup, Bengaluru.
2025 Invited research talk at the Dyson Robotics Laboratory, Imperial College London.
2025 Invited research presentation at the Citadel & Citadel Securities PhD Summit, London.

Teaching & Service

Senior/Head Tutor, ANU: Introduction to Machine Learning; Programming Fundamentals; Building Cyber-Physical Systems.
Reviewer (2021–2026): CVPR, ECCV, NeurIPS, ICML, AAAI, ICRA, IROS, IEEE T-RO and RA-L.

Skills

Programming & Scientific Computing: Python; PyTorch (Autograd, `torch.compile`/AOTAutograd); Triton; SQL; C; MATLAB; Bash; NumPy; SciPy; pandas; scikit-learn.
Computer Vision, Generative & Spatial AI: generative world models; diffusion; flow matching; 3D scene understanding; uncertainty estimation; OpenCV; Open3D; PyTorch3D; COLMAP; ROS2.
LLM/VLM Architecture & Inference: token pruning/merging/compression; FlashAttention; KV-cache optimisation; sparse/MoE routing; quantisation; speculative decoding; vLLM; hardware-aware co-design; LLM-as-judge; RAG/agent evaluation.
Quantitative Methods: statistical inference; time-series modelling; feature engineering; market microstructure; signal research; regime-aware analysis; walk-forward validation; backtesting; risk modelling; XGBoost; xLSTM; SHAP.
ML Systems & Infrastructure: mixed-precision/multi-GPU training; inference profiling; benchmarking; reproducible pipelines; Linux; Git; Docker; SLURM; CUDA; AWS (EC2, S3); Weights & Biases.
Communication & Collaboration: technical writing; research presentations; stakeholder engagement; teaching and mentoring.